Mini Paper — 2.2 Problem Solving & Programming

Time: ~35 minutes · Total: 30 marks · Answer all questions.

Mark your work against the mark scheme at the end; aim for ~1 minute per mark.

Questions

Q1. State the difference between a **function** and a **procedure**. **[2]** (AO1)

Q2. Explain what is meant by **recursion**, and describe one problem that can occur if a recursive subroutine is written incorrectly. **[4]** (AO1)

Q3. The recursive function below is called as `total(4)`.

```
function total(n)
    if n == 0 then
        return 0
    else
        return n + total(n - 1)
    endif
endfunction
```

(a) Using the call stack, trace the calls made and show the values returned as the stack unwinds. **[3]** (AO2)

(b) State the **final value** returned by `total(4)`. **[1]** (AO2)

Q4. A programmer uses the procedures below.

```
procedure changeByValue(byVal x)
    x = x + 10
endprocedure

procedure changeByReference(byRef x)
    x = x + 10
endprocedure
```

The following code runs:

```
num = 5
changeByValue(num)
print(num)
changeByReference(num)
print(num)
```

(a) State the two values that are printed, in order. **[2]** (AO2)

(b) Explain **why** the two outputs differ, referring to how the parameter is passed in each case. **[3]** (AO1)

Q5. A program to manage a library stores information about books.

Write, in **OCR pseudocode**, a class `Book` that has:

- two **private** attributes: `title` and `copies`
- a **constructor** that sets both attributes from parameters
- a procedure `borrow()` that reduces `copies` by 1 only if `copies` is greater than 0
- a **getter** `getCopies()` that returns the number of copies. **[6]** (AO3)

Q6. Define the term **encapsulation** and give **one** reason it is used in object-oriented programming. **[3]** (AO1)

Q7. A streaming service wants to analyse millions of past viewing records to discover which genres of film are often watched together, so it can recommend titles.

(a) Name the **computational method** best suited to this task. **[1]** (AO1)

(b) Justify your choice with reference to the scenario. **[2]** (AO2)

Q8. Explain the difference between a **local variable** and a **global variable**, and give **one** reason why a programmer would prefer to use local variables. **[3]** (AO1)